

# Historical Analysis and Time Series Forecasting of Stock Market using FB Prophet

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**Abstract**— Forecasting can be used in many fields such as crypto currency prediction, financial entities, supermarkets etc. We get the time series date which we use to feed the data into the algorithm is given by Yfinance with this we get refreshed data every day. The stock market prediction or forecasting helps customers and brokers get a brief view of how the market behaves for the coming years. Many models are currently in use Like Regression techniques, Long Short-Term Memory algorithm etc. FB Prophet is proven to perform better than most other Algorithms with better accuracy. From the proposed research and references we have determined Facebook's Prophet algorithm as our forecasting algorithm because it is predicting at better accuracy, low error rate, handles messy data, doesn't bother for null values and better fitting.

**Keywords**--- FB Prophet, Time series forecasting, Streamlit, Yfinance, Plotly, Matplotlib.

## I. INTRODUCTION

The technique of predicting the future share of a company's value on an exchange is known as stock market prediction. Our project aims at forecasting the future of the stock market on a specified day or over the period of time. This project will help the stock market enthusiasts to help predict the stock price and hence helping those people in getting profits. In order to do all the, operations, we need an algorithm that could help us in doing all the analysis. We have to choose an algorithm that could be flexible to holidays

and thus, helping us in analyzing seasonal data of a stock. Hence, we chose FB Prophet as our algorithm as it can be flexible to messy data and can also handle null values very well.

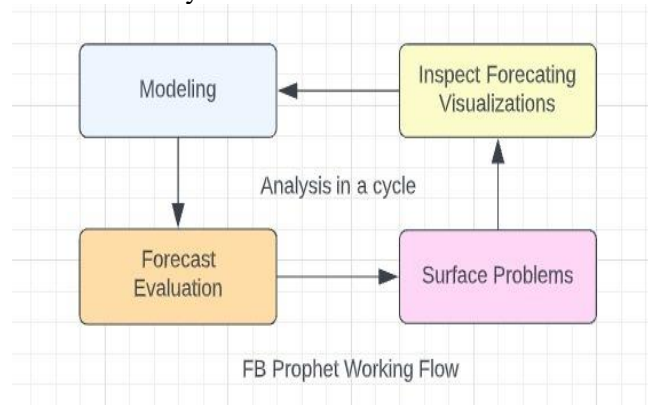


Fig. 1. Prophet's Work Flow

The major goal of our project is to project the company's share value of the future. The project aim is to help the users get a correct prediction of stock and helps him better understanding of the stock market and leads him huge profits.

The dataset we use here is from Yfinance, which provides historical data of a stock company since its inception. Data from our dataset gets refreshed every day. The ultimate goal of the proposed project was to Forecast the company's share value of the future with respect to date. We make use of

Machine learning to forecast in a better way. There are many algorithms provided, but we felt FB Prophet will do the job better. It is being used in some of the applications in Facebook. Also, we are using a web framework for python called streamlit through which we can deploy our project in public media so that anyone with an URL can view our project which we hosted in our GitHub.

## II. LITERATURE SURVEY

‘Time series forecasting model for supermarket sales using fbprophet’ by Shilpa pande, Bineet kumar jha in 2021 their paper aims in implementing the Facebook's Prophet algorithm, conducting an analysis of sales related to supermarket with provided data. fitting the obtained data into the ARIMA model for analysis.

‘Stock Price Forecasting Using Data from Yahoo Finance and Analyzing Seasonal and Nonseasonal Trend’ by Jai Jagwani, Manav Gupta in 2018. This paper offers a bigger picture of share value projection by merging the outputs of multiple analysis models in an attempt to implement a range of share value prices for brokers such that no losses are incurred.

‘Applications of Facebook’s prophet algorithm for successful sales forecasting based on real-world data’ by Kemal Korjenic, Emir zunic in 2020. This journal provides a system for reliably projecting future retail sales and classifying product offering according to predicting accuracy standards in the retail industry.

‘Time Series Model for Stock Market Prediction Utilizing Prophet’ by Neha Gupta, Dr. Lilly Sheeba S in 2021. This journal tries to highlight the importance of timing in enhancing forecasting accuracy, which is accomplished with the use of prophet algorithm. This journal uses fbprophet library to define 3 different hyper parameters namely seasonality, trend, and holidays.

‘Stock price forecasting using information from Yahoo finance and google trend’ by Selene Yue Xu in 2014. This publication aims to anticipate periodic fluctuations in share price using a combination of traditional series data analysis and data out from Yahoo.

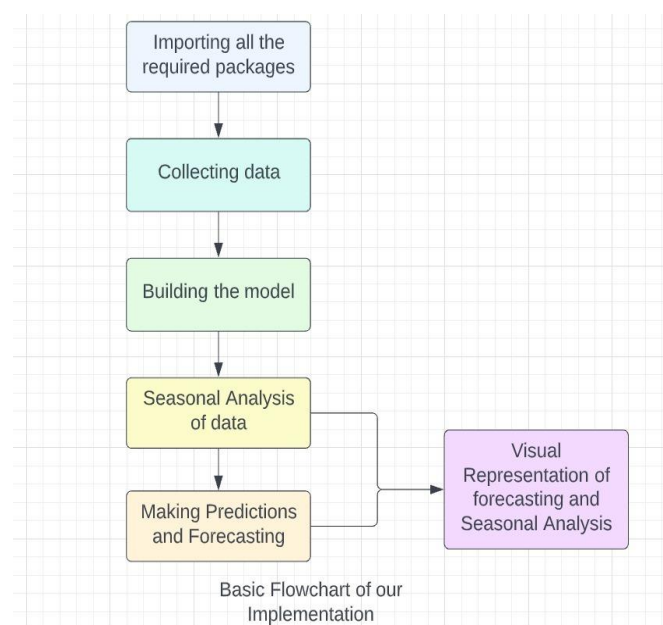
## III. RELATED WORK

Time-series prediction is one of the most important applications of deep learning. Many researches have been conducted on time-series projections using many types of algorithms such as Regression models, ARIMA, Long Short-Term Memory, Facebook's Prophet and found out that FB's Prophet provide better results. An IEEE paper released by Shilpa and Bineet for forecasting supermarket sales using FB-Prophet which is done by comparing FB Prophet and ARIMA. Their's research has shown that predictions are using an additive model of the Prophet provides results that are very nearer to actual values. Also, additionally they have added, saying that it works better for larger date sets.

## IV. METHODOLOGY

Our project could project stock market data of the future more accurately when compared to all other forecasting algorithms such as ANN, ARIMA etc.

This would be really helpful for those who are willing to invest in stocks and for the brokers too. Yfinance is a module in python, which provides daily market data using an API from Yahoo. We propose a basic flow of our project methodology with a small block diagram shown below.



**Fig. 2. Project’s Flowchart**

## A. Data Collection

To develop a better framework for our model to forecast and predict stock values we have explored many kinds of datasets in Google and Kaggle. Eventually we came to a conclusion that YFINANCE will give better data which gets refreshed every day. The data are well structured and contains every detail about a company. Our dataset contains lots of columns which are very commonly used terms for stock market analysts and enthusiasts.

1. **Date** - Historic date of every stock traded day
2. **Open** - The starting price of a company's share at the start of the day in a stock exchange market.
3. **High** - The best price at which a share was traded throughout the day.
4. **Low** - The least price at which a share was traded throughout the day.
5. **Close** - The ending price of a company's share at the end of the day in a stock exchange market.
6. **Volume** – Volume shows how many units are purchased up in a particular time period.
7. **Dividends** - A dividend is a compensation paid to a group of shareholders in the form of cash or stock.
8. **Stock splits** - A stock split is when a corporation offers additional shares to its existing owners without diminishing their stock value. It divides an organization's share to much bigger volume, which makes share's value lower and increases available volume of stocks.

The following is an example of a simple Yfinance dataset:

|   | Date                | Open     | High     | Low      | Close    | Volume  | Dividends | Stock Splits |
|---|---------------------|----------|----------|----------|----------|---------|-----------|--------------|
| 0 | 2021-11-30T00:00:00 | 115.4817 | 115.5905 | 113.9179 | 114.0960 | 8172100 | 0         | 0            |
| 1 | 2021-12-01T00:00:00 | 114.5909 | 117.7581 | 114.4227 | 114.7196 | 6287500 | 0         | 0            |
| 2 | 2021-12-02T00:00:00 | 115.0561 | 116.3922 | 114.4128 | 115.6697 | 4919800 | 0         | 0            |
| 3 | 2021-12-03T00:00:00 | 116.5308 | 117.7778 | 115.7984 | 117.6294 | 6426200 | 0         | 0            |
| 4 | 2021-12-06T00:00:00 | 118.4014 | 121.1924 | 117.9857 | 120.1037 | 8939900 | 0         | 0            |
| 5 | 2021-12-07T00:00:00 | 120.7272 | 120.8856 | 119.7573 | 120.2621 | 6123300 | 0         | 0            |
| 6 | 2021-12-08T00:00:00 | 120.7074 | 121.1825 | 119.5099 | 120.6184 | 5278100 | 0         | 0            |
| 7 | 2021-12-09T00:00:00 | 120.9549 | 123.1818 | 120.8460 | 122.8750 | 8082300 | 0         | 0            |

Fig. 3. Data collected from Yfinance

## B. Creating Our Forecasting Model

There are many forecasting models available in the market like LSTM, Regression models, ANN etc., But the newly introduced FB Prophet shows better results compared to all the other algorithms. FB Prophet is a powerful tool being used by Facebook even in its many forecasting applications. It has also been noted that this algorithm in most of the cases. Also, some of the main reasons for applying this model are this algorithm can handle messy data, can work if data contains lot of null values also provides special feature to exclude holidays.

The algorithm makes use of Pystan which is a state-of-the-art tool for statistical analysis using probabilistic algorithms.

## C. User Interface

Now, we need to implement a User Interface to deploy our application into web. Streamlit is an opensource framework in python that provides users to create and share their applications on the web. It is one the most used frameworks for data science. Using this frame work we can easily deploy our application in the public domain and host it in any cloud mediums or can be run using GitHub.

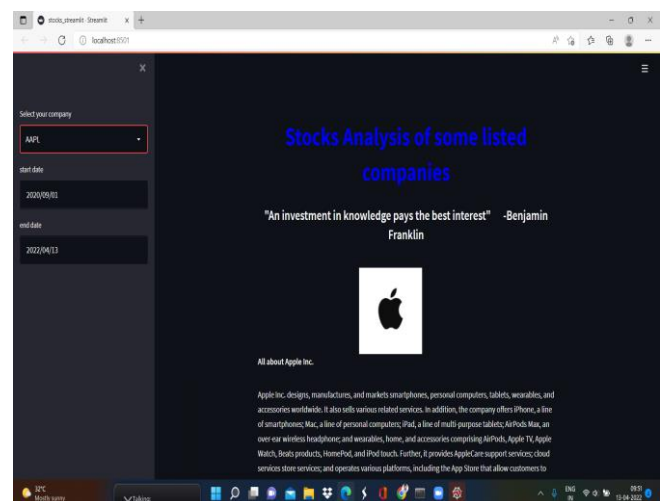
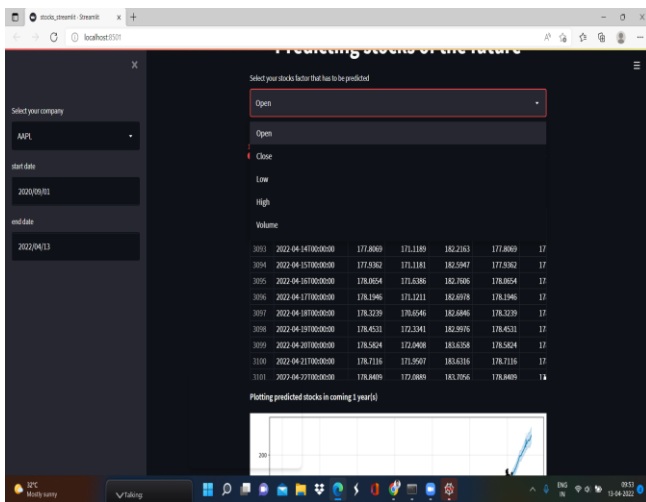
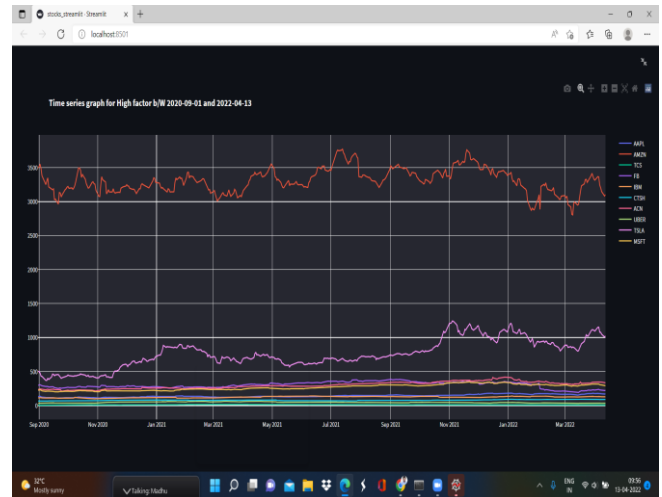


Fig. 4. Project's User Interface



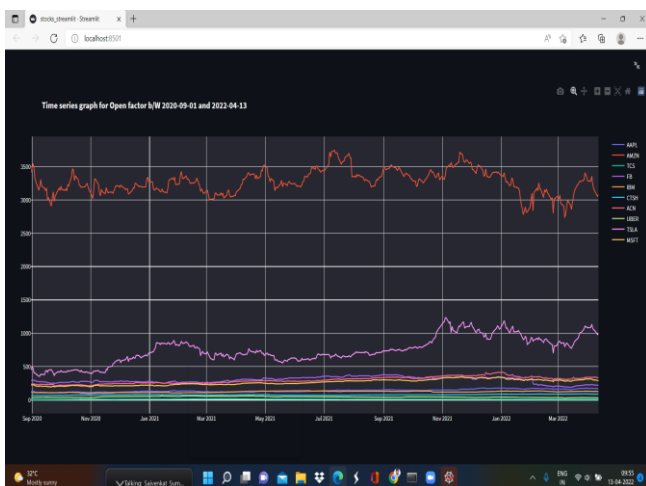
**Fig. 5. Input widgets of Streamlit**



**Fig. 7. High prices of Apple Inc. on a particular day between specified dates**

## D. Creating Interactive Plots

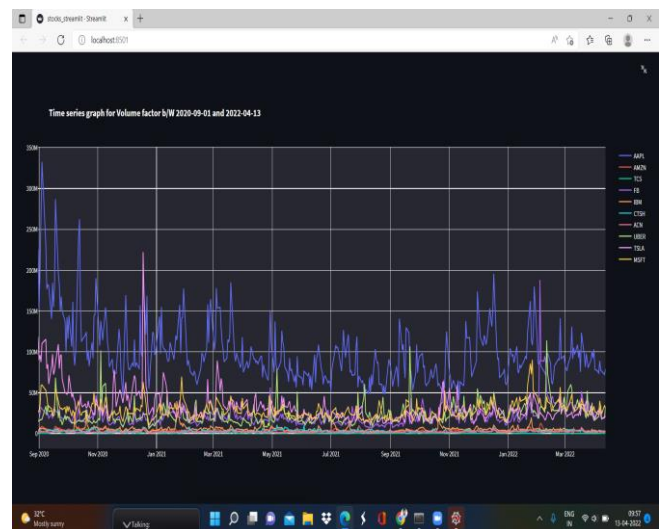
We try to implement the data we get from yfinance in several plots and graphs. The users prefer data representation in graphs and plots rather than verbal representation. So, we try to develop an interactive visualization dashboard designed to show the seasonal trends and forecasting of the stocks of the future. We are making use of several packages such as Plolty and some methods that are provided by the FB Prophet. Python allows dynamic and interactive graphs Using many packages that are already available in Python we built interactive dashboards plotting a value against a date.



**Fig. 6. Open prices of Apple Inc. on a particular day between specified dates**

Using those graphs, we could get extra information while we hover over the graph and can easily view detailed behavior by zooming in and out.

These interactive plots can be used to represent various important features of a stock industry such as open, close, high, low, dividends, etc. We implement dates on the x-axis and values on the y-axis.



**Fig. 8. Stock Volumes of Apple Inc. on a particular day between specified dates**

## E. Forecasting and Seasonal Data Analysis

Using FBprophet we try to predict the data for the coming years. We train the FB Prophet's model using the Historical data we get from Yfinance and then predict the data of the future dates. This algorithm provides results with better accuracy.



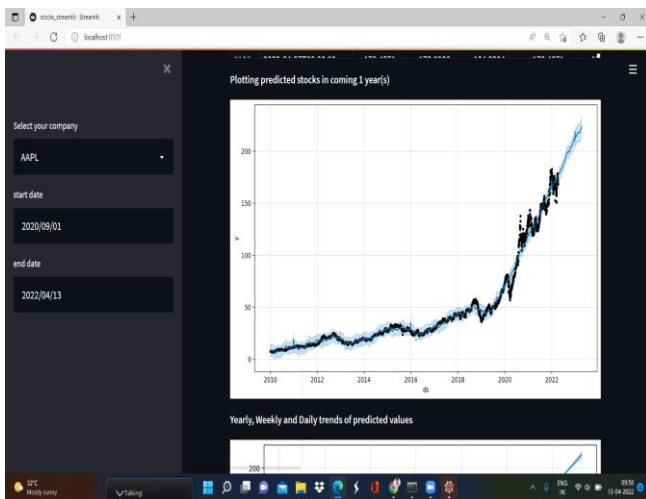


Fig. 9. Forecasting stock price for coming 1 year

Also, the algorithm works best with time series data and provides various techniques like excluding holidays and weekly, monthly, daily, yearly analysis. The algorithm also provides methods to display plots for those seasonal analysis.

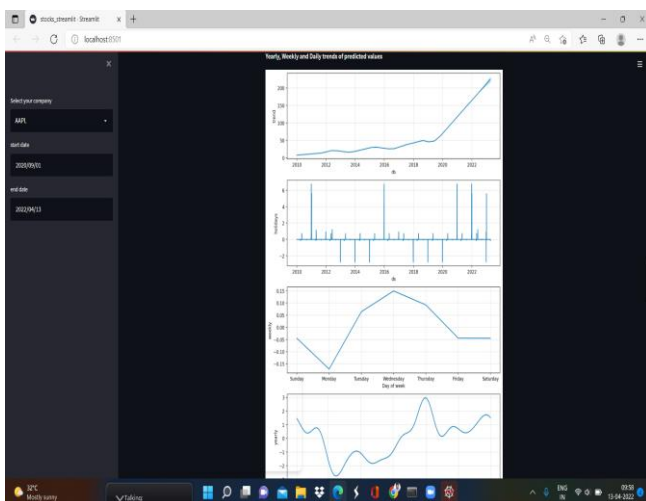


Fig. 10. Seasonal trend analysis

## F. Project Deployment

After the model has been completed successfully and user interface has been created, then we will our application cloud enable and host using using GitHub. Finally, we deploy our project in the streamlit after that streamlit provides a link with which anyone can view our projects.

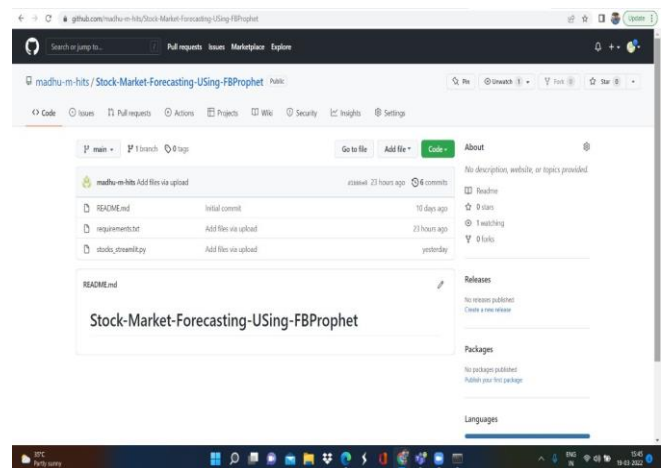


Fig. 11. Hosting in GitHub

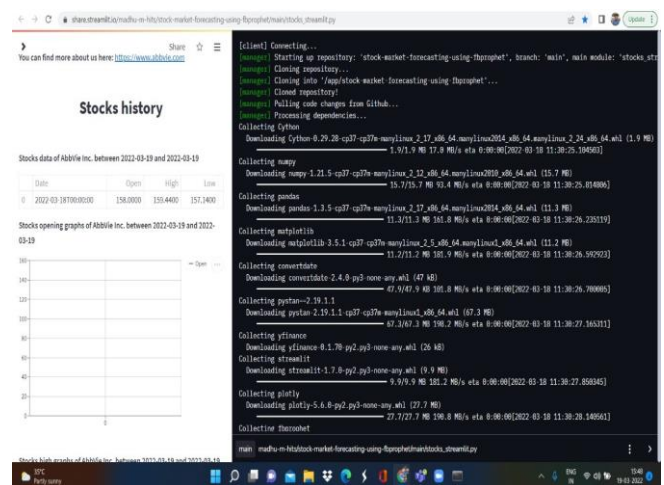


Fig. 12. Application running in Streamlit cloud

Also, our application can be shared with anyone using emails so that they can use our applications. Currently, our application is in the public domain we can also make our application for private use.

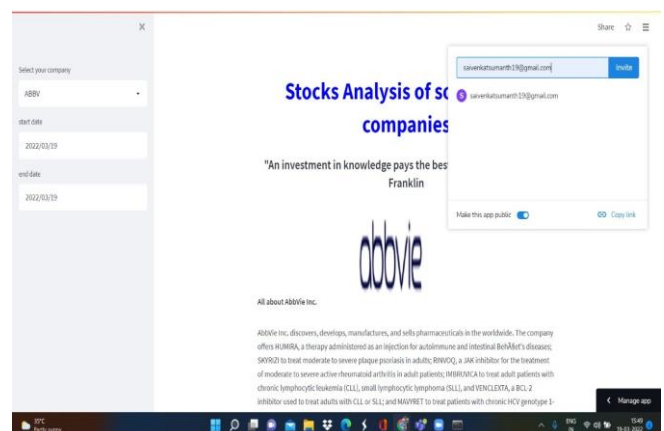


Fig. 13. Sharing URL to E-Mail

## V. CONCLUSION

We believe our project clearly aims at projecting the share value of a company of the future.

Therefore, we are aiming to implement a web-based data science application using streamlit, framework, and we try to enable our application in cloud we will host our application in github, so that anyone from anywhere can access our application using an URL.

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